Feature-Specific Transition Dynamics in Sperm Whale Codas and the CEJONG Cell-and-Tier Notation Framework

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Abstract

Sperm whale codas have been described from two angles: as combinable timing features (rhythm, tempo, ornament, and a slow inter-coda drift termed rubato), and as recurrent spectral patterns within individual clicks labeled a-vowel and i-vowel. The two have not been analyzed jointly at the level where representational decisions are made, which is the consecutive same-speaker transition. This study aligns the two systems on 1,170 vowel-labeled codas from 13 individually identified Eastern-Caribbean whales and tests whether each pair of features moves together across transitions. The resulting transition structure is feature-specific: rhythm and tempo are tightly coupled, ornament adds no additional self-transition resolution once rhythm and tempo are fixed, vowel is strongly associated with rhythm at the coda level but only weakly coupled to it at the transition level, and rubato is a relation between two codas rather than a property of one. These four findings motivate a candidate cell-and-tier notation, CEJONG, that places each feature at the level its transition dynamics suggests. The grid uses jeongganbo's vertical-time cells and hunminjeongeum's block-internal organization as architectural precedents. All transition-level claims about vowel are conditional on the labeled subset. CEJONG provides a transition-grounded representational hypothesis for downstream sequence modeling; an initial held-out encoding check supports the tier-separated choice over a fully joint cell, with broader model-based validation left open.

Keywords: sperm whale coda; animal communication; bioacoustics; autosegmental phonology; transition dynamics

1. Introduction

Sperm whales (*Physeter macrocephalus*) communicate using codas, short patterned bursts of clicks exchanged while socializing or between deep foraging dives. The traditional description (Watkins & Schevill 1977; Rendell & Whitehead 2003; Hersh et al. 2022) is a finite repertoire of around twenty Caribbean coda types, each defined by a characteristic inter-click-interval pattern. Two recent studies show that this surface description leaves substantial structure unaccounted for, and they do so from opposite directions.

Sharma et al. (2024) decompose Eastern-Caribbean (EC-1) codas into four combinable features: rhythm, tempo, an optional ornament click, and a slow duration drift across adjacent same-speaker codas they call rubato. The resulting combinatorial inventory is roughly an order of magnitude larger than the older type list. Beguš et al. (2025) approach the same signal from below the coda, finding that individual clicks fall into two recurrent spectral patterns labeled a and i.

These descriptions are complementary: every coda carries both kinds of information at once. But they have not been examined jointly at the level where annotations get bundled or separated, which is also the level where downstream sequence models inherit any structural mistakes. Marginal frequencies alone cannot determine whether two features should be bundled in the same structural unit; transition statistics provide a more direct test.

This paper aligns the two annotation systems on a joint dataset of 1,170 vowel-labeled codas from 13 individually identified whales in the EC-1 clan. For each pair of features, it asks how strongly one feature staying constant across an 8-second-adjacent transition predicts that the other does too. The resulting coupling structure is feature-specific, and each piece of it constrains representation in a specific way. CEJONG is built directly from those constraints, borrowing the vertical-time grid of *jeongganbo* and the structured-block syllable layout of *hunminjeongeum* for architecture only. Comparable architectural choices exist in autosegmental phonology and piano-roll notation; readers fluent in those can read CEJONG as one instantiation of the same general idea.

The focus is on how annotated timing and vowel-like features couple across same-speaker transitions, and on how that coupling can guide a representation for downstream sequence models. CEJONG is one such transition-grounded representation hypothesis. A held-out encoding check gives an initial internal test of this factorization, but larger labeled samples, extended vowel labels, per-individual overlays, and model-based sequence prediction remain open. All transition-level claims involving vowel are conditional on the labeled subset rather than on the full EC-1 corpus.

2. Methods

Three public files are combined: Sharma's DominicaCodas.csv and sperm-whale-dialogues.csv (Sharma et al. 2024 GitHub; EC-1: 7,770 and 3,840 codas respectively) and Beguš's codamd.csv (doi:10.17605/OSF.IO/A32KE; 1,375 hand-vowel-labeled codas from focal DTag-attached whales across 13 individuals). Beguš's labels are aligned to Sharma's tables by direct codanum identity on DominicaCodas and by a rounded acoustic key on dialogues. Cluster 17, a heterogeneous long/noise-coda cluster, is excluded because it contains unusually long codas, low vowel-label coverage, and many NOISE-labeled types outside Sharma's named coda inventory. This exclusion removes 175 dialogue rows overall and 2 vowel-labeled rows from the joined subset. The working subset is 1,170 vowel-labeled codas with 920 same-whale 8-second-adjacent transition pairs (Sharma's adjacency criterion: post-termination gap $g \leq 8$ s). Thus, 1,170 denotes vowel-labeled codas, 920 denotes vowel-labeled transition pairs, 856 denotes the validation subset after tempo-presence filtering, and 2,907 denotes all cluster-17-excluded same-whale transition pairs.

Comparisons among rhythm, tempo, and ornament use all 2,907 same-whale transitions; comparisons involving vowel use the 920 labeled pairs. Bayes factors, prior-sensitivity checks, REC-clustered bootstrap results, intra-class correlation estimates, and effective-sample-size calculations are summarized where needed below.

3. Results

Four feature-level findings (Sections 3.1–3.4) motivate CEJONG's cell and tier structure, and one layout-level observation (Section 3.5) motivates the vertical grid. Each maps to one design principle in Section 4.

Table 1. Mapping from empirical findings to CEJONG design choices.

Empirical finding	Evidence	CEJONG consequence
Rhythm-tempo couple at transitions (Section 3.1)	35-pp conditional gap, bootstrap interval [30.6, 40.0]	Bundle ⟨r, t⟩ in cell upper region
Rhythm-vowel coupling weak at transitions (Section 3.2)	Weak margin-free coupling (vowel–rhythm OR 1.40, vowel–tempo OR 1.58; CIs span 1)	Vowel on a separate lower tier
Ornament adds little transition resolution beyond ⟨r, t⟩ (Section 3.3)	3D self-transition rate equals 2D (73.8%)	Ornament at cell margin
Rubato is a pair-level relation (Section 3.4)	Defined on two-coda pairs; runs of ≥3 are linked pair chains	Run-level bracket

3.1 Rhythm and tempo move together at transitions

When a whale's rhythm does not change between two consecutive codas, its tempo stays the same 92% of the time. When the rhythm changes, the tempo stays the same only 57% of the time. The 35-percentage-point gap survives REC-clustered resampling with a tight 95% bootstrap interval of [30.6, 40.0], and it shows in the same direction in each of the four best-sampled whales shown in Figure 1A (gaps from +16 to +43 pp). This is the largest pairwise transition coupling in the dataset. Treating ⟨rhythm, tempo⟩ as a coupled bundle is therefore conservative; separating them on independent axes would lose the single largest signal here.

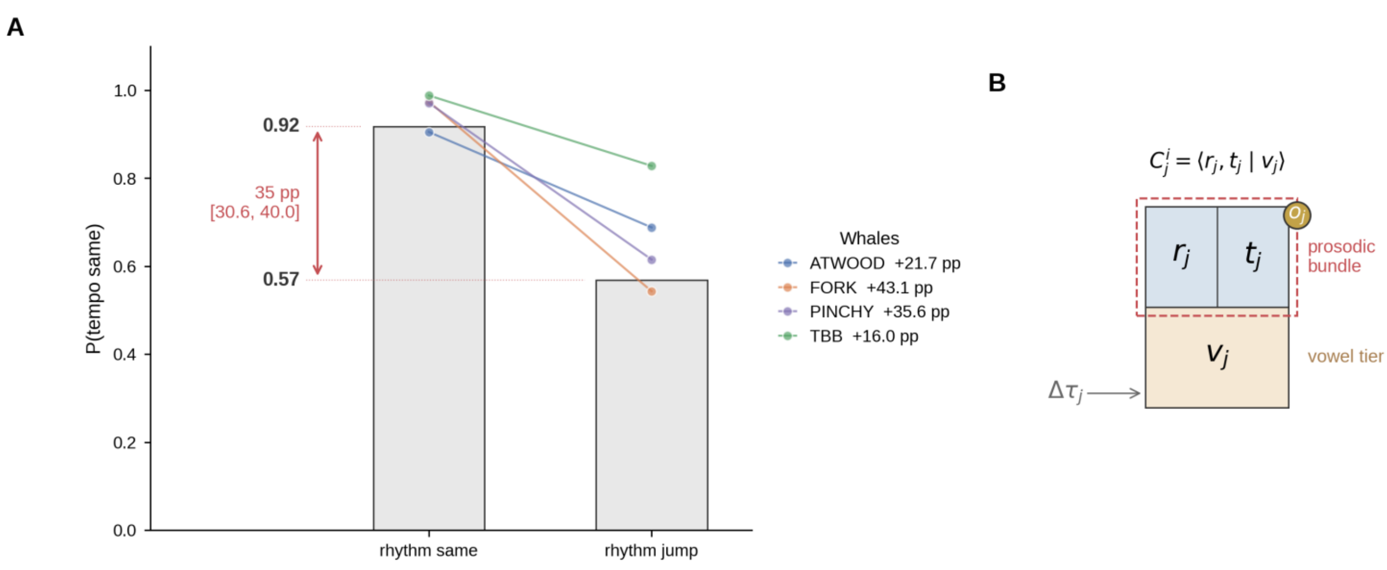

Figure 1. Rhythm-tempo transition coupling. (A) Population aggregate (gray bars): P(tempo same) drops from 0.92 to 0.57 across a rhythm change, a 35-percentage-point gap with 95% REC-clustered bootstrap interval [30.6, 40.0] (n = 2,907 transitions). Colored lines show the four best-sampled whales (ATWOOD, FORK, PINCHY, TBB), all exhibiting positive gaps in the same direction (range +16 to +43 pp). (B) The CEJONG cell bundles ⟨r, t⟩ in the upper region as the prosodic bundle (red dashed outline), with v on a separately-tracked lower tier and (o, Δτ) at the margin (full anatomy in Figure 3).

3.2 Rhythm and vowel: strong static association, weak transition coupling

This result has two opposing-looking halves, both conditional on the labeled subset.

At the coda level, rhythm and vowel are strongly associated ($\chi^2(14) = 116.28$), with the two largest rhythm classes carrying opposite vowel preferences: r2 → i and r4 → a. The direction is consistent within well-sampled individuals: r2 tends toward vowel i, whereas r4 tends toward vowel a.

At the transition level, the same two features are not robustly coupled. Across 920 same-whale transition pairs, vowel changes 6.0% of the time when the rhythm jumps and 4.5% when it stays, a gap of 1.5 percentage points with Cohen's w = 0.022. The population Bayes factor favors independence ($BF_{01} = 15.27$ under a uniform prior, range 8.50–71.40 across reasonable priors), but the REC-clustered bootstrap 95% interval is wide and crosses 1 ([0.17, 27.86]). The result is therefore reported as weak transition coupling, with subgroup and prior-sensitivity checks leaving the conclusion in the same range.

The two halves describe different quantities. The coda-level association is static: r4 codas tend to carry a, r2 codas tend to carry i. The transition-level near-null is dynamic: rhythm can change while the vowel tier usually remains stable. The two are compatible if vowel choice is set on a slower timescale than rhythm choice, perhaps tied to individual identity, social context, or behavioral state, though this dataset cannot decide among those alternatives. This distinction matters for representation: bundling vowel with rhythm and tempo inside a single cell coordinate would overstate the transition coupling, so vowel goes on a separately-tracked lower tier. The same near-null pattern recurs for the two bundle-relevant vowel comparisons. Vowel–rhythm and vowel–tempo conditional gaps are small in raw percentage-point terms, but raw gaps are descriptive because vowel changes rarely at transitions. On a margin-free odds-ratio scale, vowel remains only weakly and not robustly coupled to the timing bundle: vowel–rhythm OR = 1.40, REC-clustered 95% CI [0.28, 3.80], and vowel–tempo OR = 1.58, [0.67, 3.49]. By contrast, rhythm–tempo shows much stronger coupling (OR ≈ 8.2). Ornament-involving odds ratios are unstable here because of near-empty cells and are reported only in the derived tables. The tier decision therefore does not require proving transition-level independence. It is enough that vowel behaves unlike rhythm and tempo at the transition scale: it is highly self-stable, only weakly perturbed by rhythm and tempo changes, and performs worse when collapsed into a fully joint $\langle r,t,v \rangle$ cell in the held-out check.

3.3 Ornament adds little transition resolution beyond rhythm and tempo

Here, lower self-transition rate after adding a feature means that the added feature splits otherwise repeated cells into more specific states; if the rate does not change, the added feature contributes little additional transition resolution under this criterion. Ornament is one of the most rhythm-coupled features in the dataset at the marginal level (rhythm-ornament conditional gap +33.3 pp), but once the rhythm-tempo cell is fixed it adds no further self-transition resolution by the criterion used here. The 3D self-transition rate (rhythm × tempo × ornament) equals the 2D rate exactly, at 73.8%. This exact equality means that adding ornament to the $\langle r,t \rangle$ state does not split additional self-transition events under this criterion. The relevant hierarchy is rhythm alone at 80.4%, rhythm × tempo at 73.8%, rhythm × tempo × ornament still at 73.8%, vowel alone at 95.2%, and the full rhythm × tempo × ornament × vowel cell at 71.8%. Thus ornament adds no additional self-transition resolution once $\langle r,t \rangle$ is fixed, whereas tempo does. Ornamented and non-ornamented codas in the most ornament-rich rhythms (r16, r11, r7) also share identical click counts within each rhythm class. CEJONG accordingly puts ornament at the cell margin rather than treating it as a primary coordinate.

3.4 Rubato is a relation between codas

Sharma et al. (2024) define rubato as a duration-drift relation between two same-speaker codas satisfying click-count, start-to-start, and ornament-correction criteria. The present re-implementation reproduces Sharma's drift characterization closely (mean absolute drift 0.054 s vs 0.05 s). Restricted to runs of three or more rubato-linked codas, the EC-1 dataset contains 510 such runs, with per-individual sign profiles that differ noticeably (ATWOOD skews toward net shortening, FORK is evenly split). Rubato is therefore encoded as a run-level relation rather than as a fifth cell coordinate.

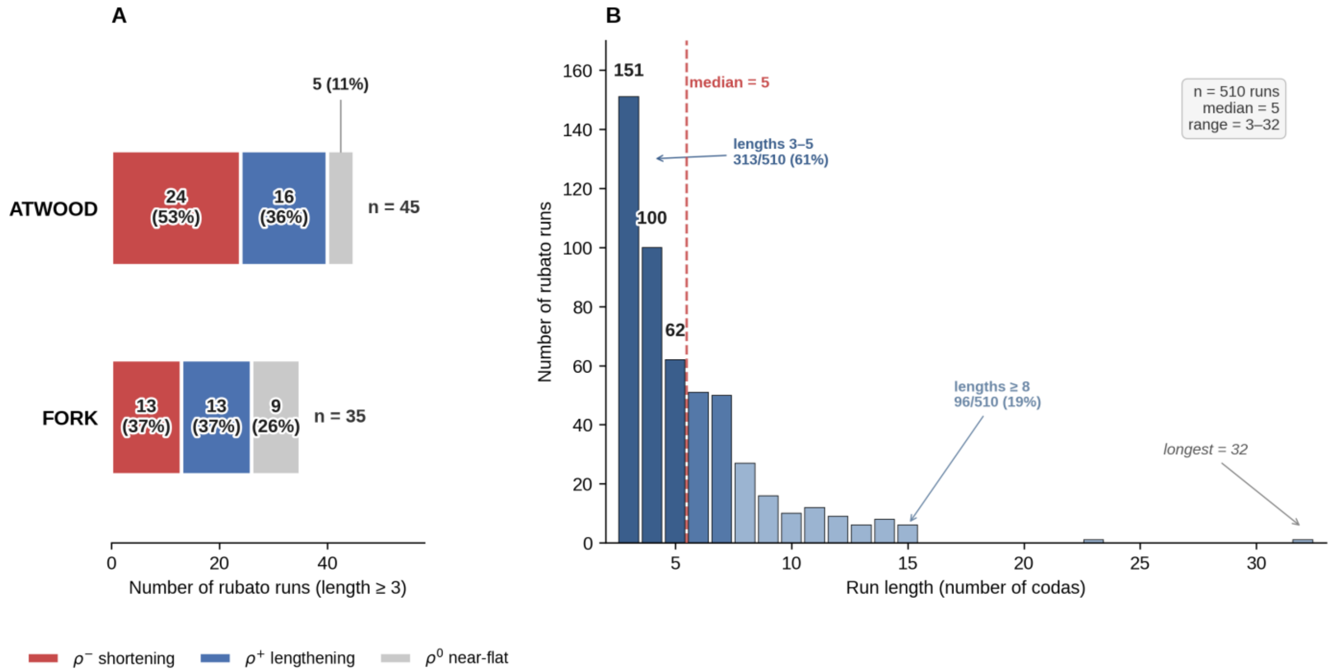

Figure 2. Rubato run profiles across 510 EC-1 runs of length ≥ 3. (A) Per-individual sign distribution for ATWOOD (n = 45) and FORK (n = 35), stacked by sign category: ρ^{-} net shortening (red), ρ^{+} net lengthening (blue), ρ^{0} near-flat (gray). Counts shown with within-individual percentage. (B) Run-length distribution; short runs dominate (151 three-coda runs, 100 four-coda runs), with the longest run spanning 32 codas.

3.5 A layout constraint

Across single-whale sequences in the EC-1 dialogues, the dominant local pattern is extended same-rhythm runs (AAA, AAAA, …) rather than ABA-style return motifs: 135 ABA triples are observed, above a 1-step Markov baseline of 63 but below a within-sequence composition-preserving shuffle expectation of 231. A vertical grid makes these runs immediately visible as a single column band; a horizontal staff buries them. This motivates the vertical-time orientation of the CEJONG grid.

4. CEJONG: a transition-grounded notation

4.1 Architectural sources

Two fifteenth-century Korean writing systems supply the architectural ideas. *Jeongganbo* is a court-music notation in which time flows vertically downward through a grid of cells, so that repetition forms a visible vertical band. *Hunminjeongeum* is a writing system in which a syllable is packaged into a structured block whose internal layout reflects featural relations among consonants and vowels rather than a linear order. CEJONG uses these systems as architectural precedents: jeongganbo supplies the vertical-time grid, and hunminjeongeum supplies the idea of a structured block whose internal layout reflects feature relations.

4.2 Design principles

CEJONG separates four kinds of information that flat labels usually conflate: cell coordinates, cell-margin annotations, transition edges, and run-level relations.

The first principle (P1) orders the grid vertically: each coda occupies one cell, and consecutive same-cell or same-rhythm cells form a visible band, exposing the AAA-style local pattern documented in Section 3.5.

The second principle (P2) treats each cell as a structured block. The upper region holds the timing form ⟨r, t⟩ as a coupled bundle, because rhythm-tempo is the largest transition coupling in the data. The lower region holds the vowel v on a separately-tracked tier, because vowel is the one feature whose transition coupling against every other feature is weak. The margin holds ornament o and the start-to-start interval Δτ, because ornament adds no additional self-transition resolution once ⟨r, t⟩ is fixed.

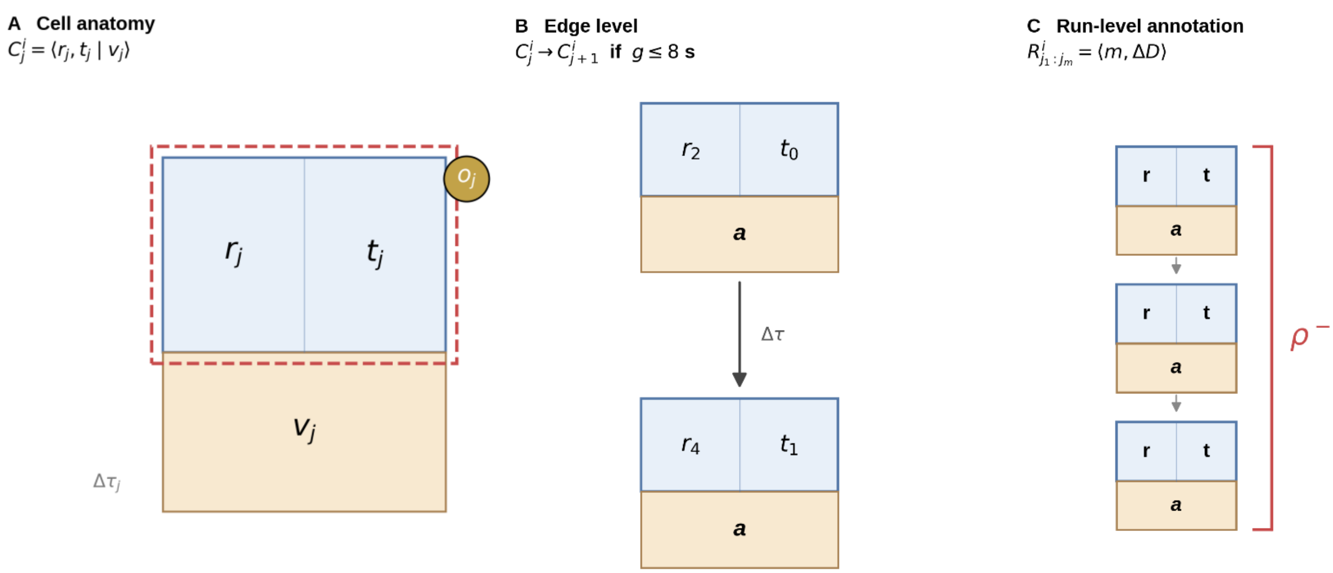

Figure 3. Anatomy of a CEJONG cell and its edge and run-level annotations. (A) Cell-internal layout ⟨r,t | v⟩ with upper rhythm-tempo bundle, lower vowel tier, and margin annotations. (B) Edge between same-individual cells linked by g ≤ 8 s. (C) Run-level bracket spanning cells satisfying the rubato criterion.

The third principle (P3) translates each within-whale 8-second-adjacency transition into a directed edge of the CEJONG transition graph; per-rhythm self-transitions appear as cell-to-cell loops.

The fourth principle (P4) allows the same cell vocabulary to support multiple per-individual layers, with each whale's edges weighted by its own observed transitions. The r4 self-transition probability spans 0.38 to 1.00 across the five best-sampled whales (Figure 4), so these per-individual differences deserve their own visible layer rather than being averaged together. Because several r4 rows are small, this range is treated as descriptive evidence for preserving per-individual layers, not as a stable estimate of each whale's transition kernel.

The fifth principle (P5) renders properties that hold across a chain of cells, such as rubato, as brackets spanning the chain with a sign marker for cumulative drift, because rubato is a pair-level relation rather than a per-coda attribute.

Formally, a CEJONG cell C_j^i for individual *i* at position *j* is

$$C_j^i = \langle r_j, t_j \mid v_j \rangle,\ A_j^i = \langle o_j, \Delta\tau_j \rangle,$$

where $r_j \in \{0, 1, \ldots, 16\}$ (rhythm cluster), $t_j \in \{0, 1, 2, 3, 4\}$ (tempo class), $v_j \in \{a, i\}$ (vowel, when observed), $o_j \in \{0, 1\}$ (ornament), and $\Delta\tau_j$ is the start-to-start interval to the previous cell. A within-whale edge between C_j^i and C_{j+1}^i exists when the post-termination gap $g_j \leq 8$ s. A rubato run is a maximal chain of $m \geq 3$ same-individual cells in which every consecutive pair satisfies Sharma's rubato criterion; each run carries $R = \langle m, \Delta D \rangle$ with sign markers ρ^-, ρ^+, and ρ^0 for net shortening ($\Delta D > +0.01$ s), net lengthening ($\Delta D < -0.01$ s), and near-flat drift ($|\Delta D| \leq 0.01$ s) respectively.

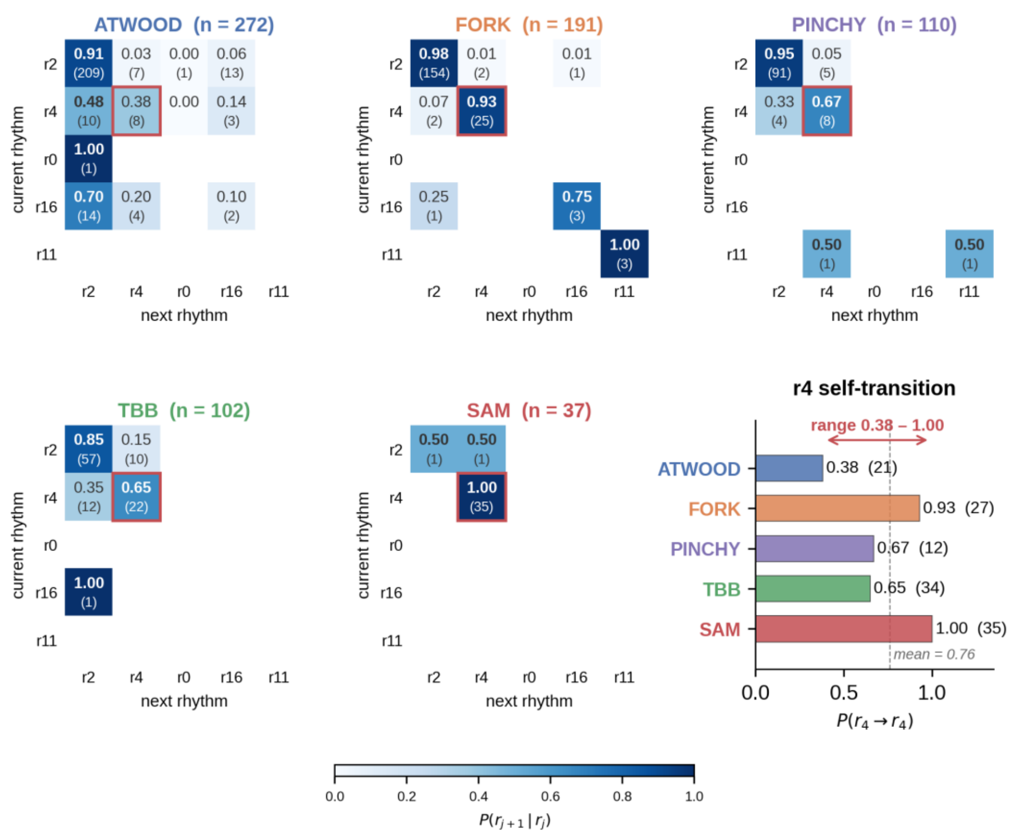

Figure 4. Per-individual rhythm transition kernels for the five best-sampled whales. Heatmaps show row-normalized $P(r_{j+1} \mid r_j)$ with raw counts; the r4 self-transition cell is highlighted. The r4 self-transition probability ranges from 0.38 in ATWOOD to 1.00 in SAM; this descriptive spread motivates CEJONG's per-individual overlay layer on a shared cell vocabulary.

4.3 Worked example: ATWOOD

A 12-coda ATWOOD excerpt (REC sw061b001_4333, ~61 s) spans three rhythm classes (r2, r4, r16) on a uniform vowel tier v = a. The excerpt contains two rubato runs with opposite signs: a net-shortening run (ρ^{-}) over cells 1–5 and a net-lengthening run (ρ^{+}) over cells 8–12. The boundary between cells 5 and 6 has a gap of 9.46 s and is excluded from edge counts. The excerpt visualizes three principles at once: same-rhythm visual recurrence (P1, in the r4 cells 5–7), opposite-signed rubato runs (P5), and individual return-state behavior (P4, with r2 as ATWOOD's preferred return state).

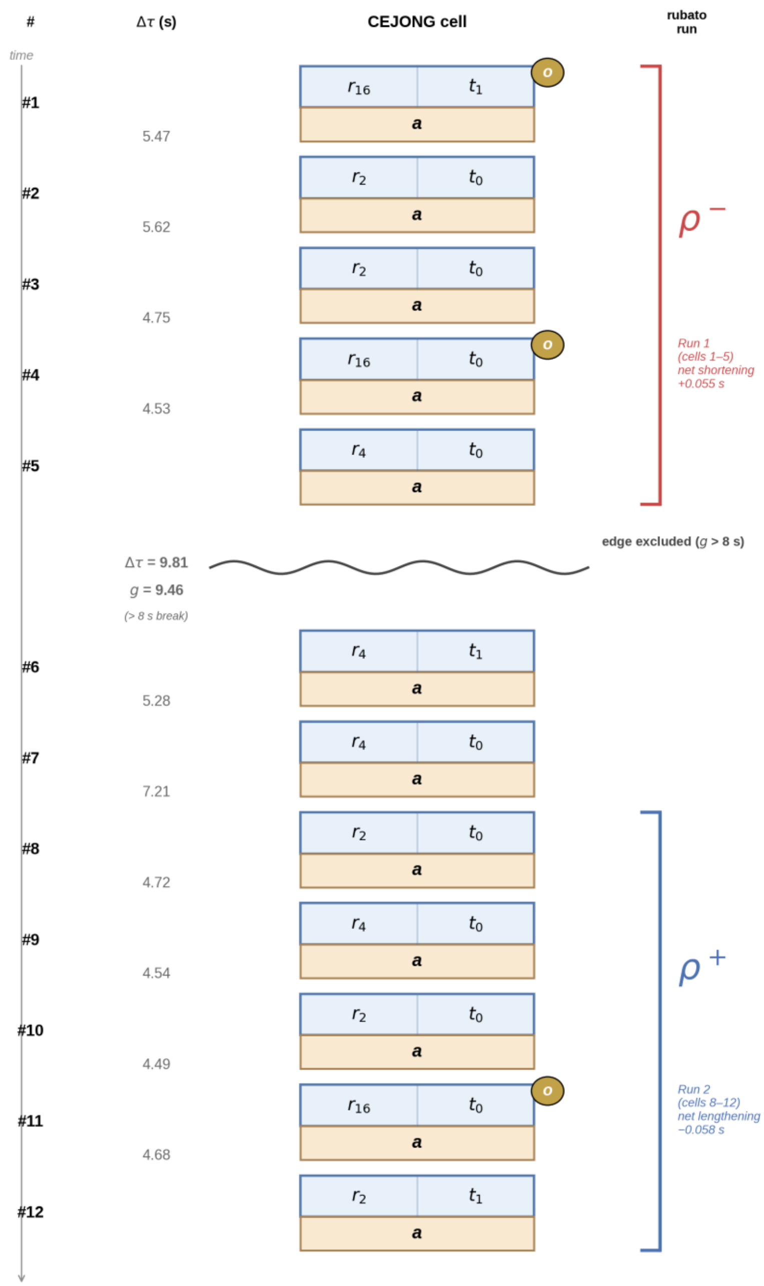

Figure 5. ATWOOD same-speaker excerpt, REC sw061b001_4333, 12 codas, ~61 s. Right-side brackets mark Run 1 (cells 1–5, ρ^{-} net shortening, red) and Run 2 (cells 8–12, ρ^{+} net lengthening, blue). The cell 5 → 6 boundary (g = 9.46 s) is excluded from edge counts.

4.4 Worked example: FORK

A 10-coda FORK excerpt (REC sw085a003_20590, ~39 s) contrasts with the ATWOOD excerpt. All three streak brackets here mark identical-cell streaks: the full ⟨r, t, v⟩ tuple is constant inside each bracket. Cell 3 is a single-occurrence rare-cluster ornamented cell. Cell 7 is the moment all three layers shift at once, with the vowel switching from a to i alongside rhythm and tempo. This is a local instance of rhythm-vowel co-change, in contrast to the population background where rhythm-vowel transition coupling is not robustly detectable on average. It is used as a worked example of the notation, rather than as evidence that such co-change is typical.

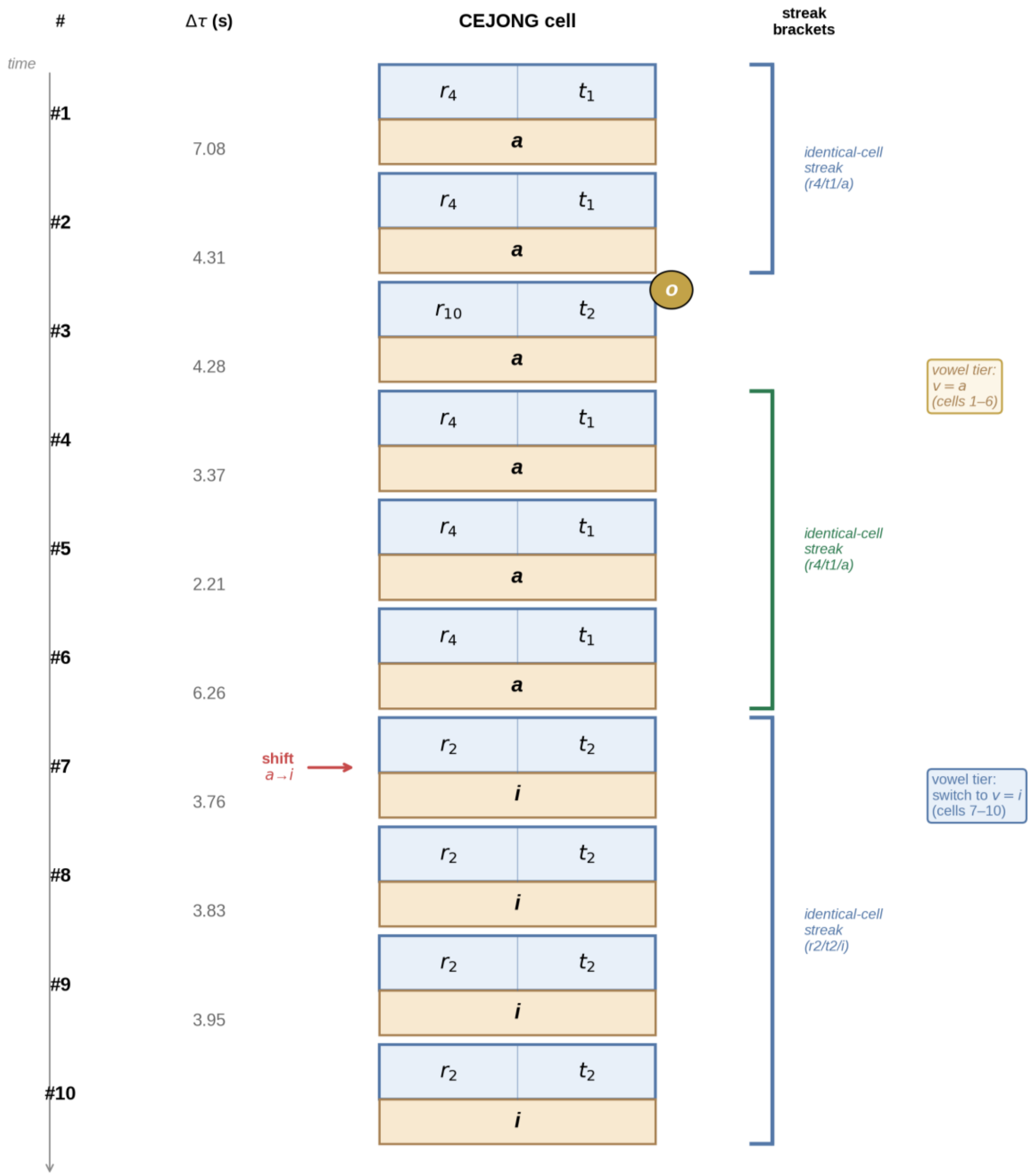

Figure 6. FORK same-speaker excerpt, REC sw085a003_20590, 10 codas, ~39 s. Three identical-cell streaks (cells 1–2, 4–6, 7–10); rare-cluster ornamented cell at 3; simultaneous r, t, v shift at cell 7 marked with a red arrow.

4.5 Scope and related notations

Current CEJONG encodes coda-level timing features, the vowel tier, same-whale transition edges, and run-level rubato. Extensions to sub-coda dynamics, cross-whale chorus structure, and richer behavioral context would require additional layers. Beguš et al. (2026) report finer phonological structure within the vowel system, and extending v from {a, i} to {a, i, ī} with a sub-coda onset annotation fits inside the existing framework.

The closest direct precedent is Davitt et al. (2026), who transcribe codas in Western music notation requiring beat interpretation. CEJONG uses a structured cell for Sharma's four features, integrates Beguš's vowel labels into the same representation, and keeps rubato at the run level rather than translating codas into beat-based Western notation. The Sperm Whale Phonetic Alphabet of Sharma et al. (2024) is extended here with the vowel tier inside each cell, rubato moved to the run level, and same-whale and cross-whale layers kept separate.

4.6 Held-out encoding check

As a downstream check of the CEJONG factorization, simple empirical transition-table encodings were compared under REC-clustered 10-fold cross-validation. On the 856-pair vowel-labeled validation subset, after validation-specific exclusions, the CEJONG-style BUNDLE+V encoding has the lowest mean held-out NLL, where lower NLL indicates better prediction, and outperforms the fully joint ⟨r,t,v⟩ cell in 9 of 10 folds. Its advantage over FLAT is positive but weaker. On the full 2,907-pair rhythm-tempo transition set, bundling ⟨r,t⟩ outperforms the flat rhythm/tempo encoding in all 10 folds. The result gives an internal held-out check on the factorization motivated by the transition analyses above, especially the choice to keep vowel separate from the fully joint ⟨r,t,v⟩ cell.

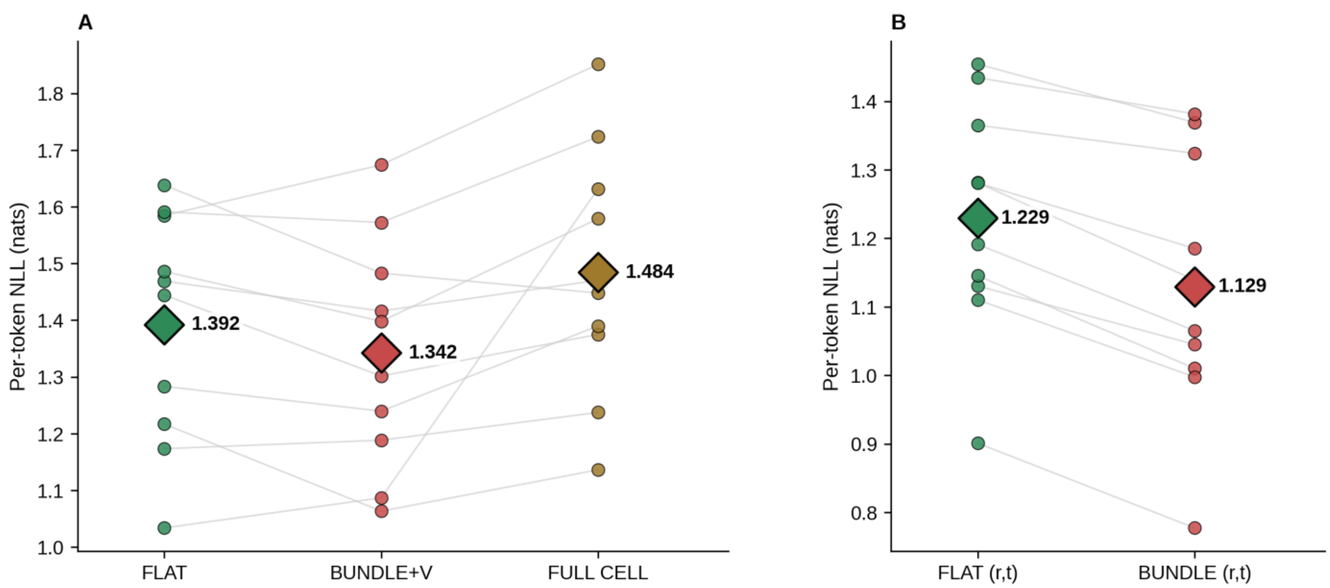

Figure 7. Held-out cross-entropy check of CEJONG encoding choices. Each circle is one REC-clustered cross-validation fold, lines connect the same fold across encodings, and diamonds mark cross-fold means. (A) Vowel-labeled validation subset after validation-specific exclusions (n = 856 transition pairs). FLAT factorizes rhythm, tempo, and vowel as independent channels; BUNDLE+V treats ⟨r,t⟩ as a joint state with vowel on a separate tier; FULL CELL joins all three as ⟨r,t,v⟩. BUNDLE+V outperforms FULL CELL in 9/10 folds (Wilcoxon $p = 0.002$), while its advantage over FLAT is weaker (7/10 folds, $p = 0.08$). (B) Full rhythm-tempo transition set (n = 2,907, no vowel filter). BUNDLE (r,t) outperforms FLAT (r,t) in 10/10 folds (Wilcoxon $p = 0.001$).

5. Discussion

Sharma et al. (2024) and Beguš et al. (2025) describe sperm whale codas from two angles, and when those descriptions are placed side by side on the same dataset, the features do not all behave as if they belong at the same structural level. Rhythm and tempo move together; ornament adds little transition resolution beyond the rhythm-tempo cell; vowel appears to operate on a slower timescale than rhythm; rubato is a relation between two codas. CEJONG uses this pattern to assign these features to different representational levels: a cell-internal timing bundle, a lower vowel tier, the cell margin, and run-level annotation.

The rhythm-vowel result has two halves that look contradictory. The static coda-level association and the dynamic transition-level near-null are compatible if each whale stays near its preferred (rhythm, vowel) combinations on a slow timescale while rhythm cycles on a faster one. A per-rhythm Cochran-Mantel-Haenszel analysis shows that the within-individual direction is consistent across well-sampled whales. The per-individual differences concentrate at r4, a rhythm class in the EC-1 clan-identity-associated region of the repertoire (Hersh et al. 2022; Sharma et al. 2024), making variation at this vertex a useful empirical handle on individual structure.

Several limitations qualify these claims. The vowel-labeled subset is 1,170 codas across 13 individuals, and Beguš's hand labeling was performed on focal DTag-attached recordings that favor specific individuals and behavioral contexts. All transition-level claims involving vowel are therefore conditional on this subset. CEJONG also inherits Sharma et al.'s 18 rhythm clusters and 5 tempo classes as working annotations rather than re-estimating those boundaries. For the rhythm-vowel transition test, BF_{01} is above 10 across all reasonable priors at the population point estimate, but the REC-clustered bootstrap interval is wide and crosses 1, and the design effect of 1.40 reduces the effective sample size to about 659. A cross-whale matching reanalysis shows that REC structure can explain apparent inter-speaker matching effects as well.

A parallel geometric reanalysis (Bond 2026) embeds the same DSWP coda taxonomy in a continuous hyperbolic space and confirms several token-level universals (Menzerath, Zipf, higher-order Markov structure, turn-taking). That framework operates at a different level from CEJONG: it embeds the existing coda type taxonomy, while CEJONG derives a discrete cell-and-tier notation from transition-level couplings on the joined subset. Bond's finding that rhythm classes carry systematically different topological signatures is consistent with CEJONG placing rhythm inside the upper cell region. Whether CEJONG-encoded sequences also preserve those token-level universals is not tested here.

CEJONG functions as an encoding hypothesis between raw labels and downstream sequence models. The held-out check in Section 4.6 supports tier separation most clearly, while broader model-based predictive tests are left open.

6. Conclusion

Four feature-level transition findings on a 1,170-coda joint Sharma–Beguš subset suggest a representational geometry in which rhythm and tempo belong inside one structural unit, vowel on a separate tier, ornament at the cell margin, and rubato at the chain level. CEJONG, motivated by these findings and architecturally inspired by jeongganbo and hunminjeongeum, places each feature at the level its transition dynamics suggests. A held-out encoding check, summarized in Figure 7, supports the tier-separated BUNDLE+V encoding over the fully joint $\langle r,t,v \rangle$ cell; extensions include a larger vowel alphabet, behavioral conditioning, and per-individual overlays. Larger per-individual samples in the DSWP archive beyond the labeled subset would strengthen both the kernel comparison and the rubato direction profile. Conditioning CEJONG on behavioral state (Andreas et al. 2024), constructing transition graphs per clan (Hersh et al. 2022), and extending the vowel alphabet to {a, i, ī} (Beguš et al. 2026) all fit inside the same framework.

Data and code availability

Sharma DominicaCodas.csv and sperm-whale-dialogues.csv are available at github.com/pratyushasharma/sw-combinatoriality. Beguš codamd.csv is available on OSF (doi:10.17605/OSF.IO/A32KE). The processing scripts, reproduction notebook, figure-generation code, cross-entropy validation package, derived tables, odds-ratio summaries, robustness notes, worked-example source rows, and additional rendered figures are available at *https://osf.io/ctnfv/*. This study reanalyzes previously published public datasets and involves no new animal handling, field collection, or experiments.

Competing interests

The author declares no competing interests.

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